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Advanced Scale Removal in a Non-Injective Well Using Custom Jetting Tools and Chelating Chemistry

Mohamed Adel A. Mohamed, Mohamed Abdulla Al Bloushi, Mahra Jalal Albloushi, Mohamed Ibrahim Al Janahi, Mohamed H. Alali, Mohammed E. Mohamed, Sabha Alqaydi, Punj Sidharth, Mayank Varshney, Marinus Grutters, Yugal Kishore Ma heshwari, Mohamed Masud Akhtar, Bilal Ahmad, and Steven Halverson, Abu Dhabi National Oil Company, Abu Dhabi City, Abu Dhabi, United Arab Emirates; Charbel Ramy, Superior Abu Dhabi Company LLC, Abu Dhabi, United Arab Emirates

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Abstract

This paper presents a comprehensive case study of a complex hydro-mechanical intervention executed in gulf region for a major oil and gas operator on injection well in which tubing access was completely obstructed due to severe scale deposition. The blockage extended from surface with extensive deposits from surface to end of tubing, preventing the deployment of any conventional intervention methods such as wireline, mechanical setting tools, or gauges. The scale acted as a physical barrier, completely halting injectivity and compromising the well's integrity and functionality.

Initial diagnostics included downhole scale sampling, which was analyzed in contractor and operator laboratory using techniques such as X-ray diffraction (XRD), scanning electron microscopy (SEM), and inductively coupled plasma (ICP) analysis. Results confirmed that the scale was primarily composed of barium sulfate (BaSO_4) and strontium sulfate (SrSO_4), both highly insoluble sulfates known for their extreme hardness and chemical resistance. Additionally, the analysis identified a significant presence of iron-based compounds, which increased the tenacity and structural complexity of the deposit, further reducing the effectiveness of conventional acid treatments. To address this challenge, a multi-phase strategy was adopted. A chelating agent based on EDTA (ethylenediaminetetraacetic acid) was formulated and tailored to the specific solubility characteristics of the scale. This formulation was enhanced with a specialized surfactant package to reduce interfacial tension and alter wettability, enabling deeper chemical penetration and improved reactivity with the dense scale layers. Laboratory solubility tests validated the dissolution performance of the formulation under simulated downhole conditions, including high temperature and pressure.

In field application, the chemical treatment was pumped using a Coiled Tubing with high-rate displacement approach to improve contact efficiency and reaction kinetics. Additionally, a custom-engineered, flow-activated, self-rotating jetting tool was deployed. This tool featured four nozzles, two facing downward to disrupt and perforate scale bridges, and two directed upward to lift and displace

dislodged debris, combined with a rotating head to avoid the risk of sticking. After a 48-hour chemical soak, mechanical jetting was performed, which successfully disintegrated and flushed out the softened scale. Due to the high iron content, a follow-up batch of inhibited acid was injected to target and dissolve residual iron-rich components. A subsequent post-treatment run verified full tubing accessibility and complete clearance along the entire length of the previously obstructed interval.

This intervention demonstrates a successful translation of laboratory-scale analysis and solubility modelling into field-scale application. It provides a validated, replicable workflow for restoring access in wells blocked by complex multi-phase scale, using a combination of engineered chemical systems and mechanical jetting solutions.

Introduction

Scale deposition remains one of the most persistent and technically challenging flow assurance problems in oil and gas production, particularly in mature reservoirs or wells exposed to incompatible brine chemistries and fluctuating thermodynamic conditions [1]. Among the various scale types encountered, barium sulphate (BaSO_4) and strontium sulphate (SrSO_4) are notorious for their extreme chemical resistance, low solubility, and mechanical hardness, making them exceptionally difficult to remediate using conventional intervention techniques. When such deposits accumulate within the production tubing, they can significantly impair or entirely block well access, jeopardizing not only hydrocarbon flow but also critical safety functionalities such as the deployment and setting of Surface-Controlled Subsurface Safety Valves (SSS-Vs) or blocking production tubing and disabling any well intervention due to scale obstruction.

This paper presents an in-depth case study of a complex hydro-mechanical and chemical intervention performed on a well suffering from complete tubing blockage due to massive scale deposition. The obstruction extended from surface with extensive barriers down to the downhole nipple profile and Wireline Entry Guide, keeping traditional intervention tools including wireline, coiled tubing-conveyed mechanical setting tools, and downhole gauges, ineffective or completely non-deployable. Initial production diagnostics and well integrity assessments confirmed zero injectivity and no mechanical access, prompting the need for a non-conventional, multidisciplinary approach. Laboratory investigations, using advanced analytical techniques characterizing the scale as predominantly BaSO_4 and SrSO_4 , compounded by a considerable fraction of iron-rich phases [2]. The presence of iron oxides and hydroxides not only increased the deposit's structural complexity but also rendered traditional acid-based solutions ineffective, requiring a tailored formulation to target both the sulphate and iron-based components.

To address this, a multi-stage treatment workflow was developed combining chemically engineered solutions with customized mechanical jetting technology. A high-performance EDTA-based chelating formulation, improved with surfactants for enhanced wettability and penetration, was lab-validated under simulated downhole conditions [3]. In the field, the solution was deployed through Coiled Tubing (CT) using a high-rate displacement strategy to improve scale contact time and reaction kinetics. Following a batch of inhibited hydrochloric acid (HCl) was deployed to solubilize iron-rich residues and acid-soluble components. Complementing the chemical phase, a flow-activated, self-rotating jetting tool was used to physically disrupt and dislodge the softened scale matrix. This comprehensive intervention resulted in full restoration of tubing access, successful re-setting of the SSS-V, and full recovery of well injectivity. The case exemplifies how integrated chemical and mechanical strategies, guided by robust laboratory characterization, can offer effective and scalable solutions to otherwise intractable flow assurance and integrity issues. Moreover, it introduces a replicable methodology for addressing multi-phase scale deposits in high-risk, high-value wells.

Objective

The core objective of the intervention was to restore full tubing access across a severely scaled interval from surface down to Wireline Entry Guide (WLEG), where conventional intervention methods had failed due to a complete obstruction caused by a tenacious, multi-phase scale. This blockage stopped injectivity and prevented critical operations, including the setting of the Surface-Controlled Subsurface Safety Valve (SSS-V) and the deployment of diagnostic tools. To precisely define the scale composition and inform the design of an effective treatment, scale samples were extracted and analyzed at two independent laboratories to ensure accuracy, repeatability, and analytical integrity. A suite of advanced diagnostic techniques including X-ray Diffraction (XRD) for crystallographic profiling, Scanning Electron Microscopy (SEM) for morphological and structural analysis, and Inductively Coupled Plasma (ICP) spectroscopy for elemental composition confirmed the scale to be primarily composed of barium sulphate (BaSO_4) and strontium sulphate (SrSO_4), with substantial quantities of iron-based compounds that increased matrix rigidity and chemical resistance. Building on this diagnosis, a systematic solubility study was conducted under simulated downhole conditions (temperature, pressure, and fluid environment) to screen and optimize chemical dissolution strategies. This led to the formulation of a custom EDTA-based chelating system, engineered for enhanced performance through the addition of specialty surfactants that improved wettability, reduced interfacial tension, and promoted deeper penetration into the dense scale matrix [4]. The operation aimed to chemically dissolve the scale and physically dislodge residual deposits, enabling complete mechanical access for intervention tools and the reinstallation of the SSS-V.

Method

To design an effective chemical intervention strategy for dissolving the highly resistant scale obstructing the wellbore, an extensive laboratory testing program was executed in two independent facilities to ensure analytical rigor and cross-verification. The workflow began with the physical collection and preservation of scale samples under nitrogen-purged containers to prevent oxidation prior to analysis. The first phase of testing focused on mineralogical and elemental characterization of the scale. For phase identification, solubility tests were conducted using a conventional laboratory method, operating at room temperature and high temperature to meet real-case conditions. The results confirmed the presence of crystalline barium sulphate (BaSO_4) and strontium sulphate (SrSO_4) as dominant phases. Additionally, deposit revealed a highly compacted, layered morphology with minimal porosity and a robust structure composed of interlocked sulphate crystals, heavily encrusted with iron oxides (Fe_2O_3 , Fe_3O_4) as demonstrated in the [Table 1](#) [5].

Table 1—Scale analysis including composition and solubility

Analysis of Prepared Sample Residue			
Acid Soluble material	weight %	Acid insoluble material	weight %
Na	0	Si	0
Mg	0.48	Sr	23.29
Al	0.04	Ba	25.94
P	0.27		
S	14.98		
Cl	0.67		
K	0		
Ca	1.21		
Cr	0		
Mn	0.18		
Fe	24.73		
Co	0		
Ni	0		
Cu	0		
Zn	0		
Mo	0.66		
Cd	0		
Pb	0		
Ga	0		
Sc	0		
Ti	0.29		
V	0		
Sb	0		
W	0		

Quantitative elemental profiling was achieved confirming high levels of Ba^{2+} , Sr^{2+} , Fe^{3+} , and minor traces of Ca, Mg, and Si, providing a detailed chemical signature of the deposit as stated in [table 1](#).

A comprehensive elemental analysis of the scale deposits clogging the production tubing was conducted using X-ray fluorescence (XRF), focusing on acid-soluble components. The findings revealed that the predominant constituents of the deposit are iron (Fe) and sulphur (S), with respective weight percentages of 24.73% and 14.98% as mentioned in [figure 1](#). These two elements strongly indicate the formation of iron sulfide (FeS) or similar iron-sulfur-based compounds, which are commonly encountered in production systems due to corrosion processes, reactions with hydrogen sulfide (H_2S), and microbial activity. The presence of iron at such a high concentration suggests severe internal corrosion, likely stemming from contact between steel surfaces and corrosive fluids under sour conditions.

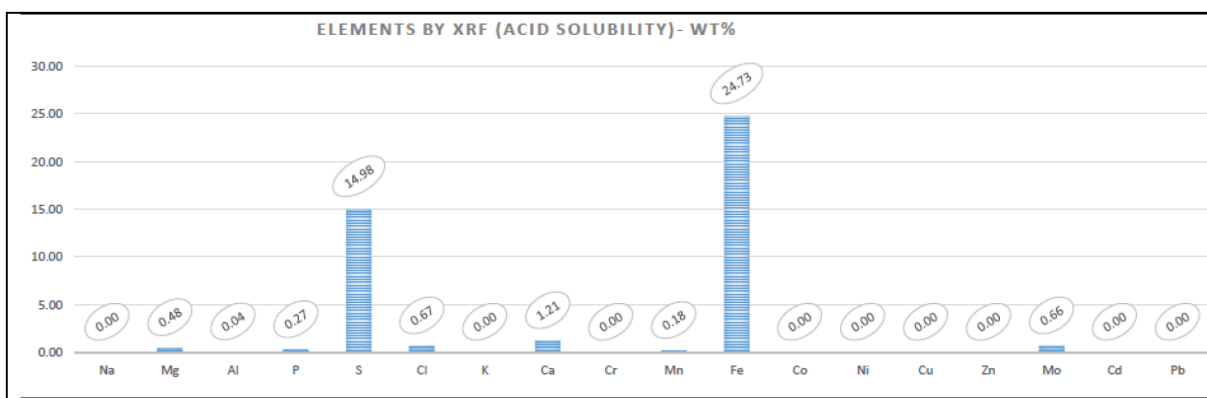
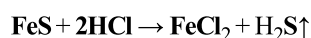


Figure 1—elemental analysis of scale deposit revealing its iron and sulphur content

The elemental distribution also includes trace levels of magnesium (0.48%), calcium (1.21%), and molybdenum (0.66%), among others, but these are relatively minor contributors compared to the overwhelming dominance of Fe and S. This compositional profile not only identifies the type of scale but also informs the choice of chemical treatment for its dissolution.

To dissolve the FeS-rich scale, an HCl-based acid system is typically employed. Hydrochloric acid is effective in breaking down iron-based scales through the conversion of iron sulfide into soluble iron chloride and hydrogen sulfide gas, following the reaction:



This reaction effectively liberates the iron and sulfur components from the tubing surface. However, the treatment must be handled with caution due to the evolution of toxic and flammable H_2S gas. Therefore, appropriate scavengers (e.g., triazines) are often added to the treatment recipe to capture and neutralize the released H_2S . Additionally, corrosion inhibitors are critical in the acid system to prevent further attack on the steel tubing, especially when using strong acids like HCl in high concentrations (typically 15–28%).

For more stubborn or passivated iron sulfide scales, blended systems such as HCl-acetic acid or HCl-formic acid may be used to enhance dissolution efficiency while controlling reaction rates and reducing gas evolution. Formulated chelating agents such as EDTA was proposed to be introduced pre-HCl flush to chelate residual iron ions and help prevent re-precipitation [6]. Furthermore, Nano surfactants may be incorporated to improve wettability and acid contact, especially in the presence of oil-wet surfaces.

The treatment is usually executed through coiled tubing with jetting tools or bullheading methods, ensuring intimate contact between the acid and scale. Operational parameters such as temperature, contact time, and flow rate must be optimized to maximize dissolution while minimizing corrosion and secondary damage. The effectiveness of the operation is verified through pre- and post-job flow measurements, visual inspection (if feasible), and well accessibility performance.

In conclusion, the XRF results distinctly identify iron sulfide as the dominant scale constituent. An HCl-based acid system, enhanced with scavengers, inhibitors, and possibly formulated chelating agents combined with nano-surfactant, is the most appropriate and effective method for dissolving such deposits and restoring well productivity.

Following the previous analysis, a detailed solubility test was performed on the scale material extracted from clogged production tubing, providing valuable insight into its complex chemical composition and informing the design of a targeted chemical treatment. The pie chart analysis reveals that the scale comprises a mixed matrix of organic and inorganic components, distributed across four solubility classes. The breakdown is as follows: a significant portion of the scale is identified as inorganic acid-soluble matter, which suggests the presence of components such as iron sulfide (FeS), calcium carbonate (CaCO_3), or iron oxides, which can be effectively dissolved using strong mineral acids such as hydrochloric acid (HCl).

In addition, another considerable fraction is inorganic acid-insoluble matter, typically indicating more refractory phases such as silica, iron silicates, or metal sulfates (e.g., barium sulfate), which are resistant to traditional acid-based systems and require chelating agents or mechanical disruption for removal as demonstrated in figure 2.

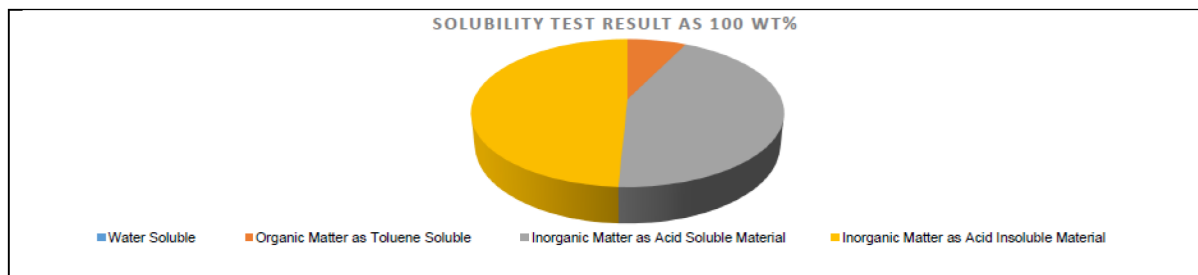


Figure 2—diagram describing the organic and inorganic content of scale sample

Furthermore, a measurable component of the scale was categorized as organic matter soluble in toluene, indicating the presence of hydrocarbon-based residues, possibly asphaltenes or paraffinic waxes that have co-precipitated with the inorganic matrix. The coexistence of these different types of materials, particularly the combination of organic and inorganic acid-insoluble components, presents a major challenge to conventional single-phase treatments and strongly justifies the need for a hybrid formulation.

Laboratory experimental

Based on these findings, a comprehensive chemical treatment system was engineered, combining chelating agents and surfactant packages to target the resistant components of the scale. Chelating agents such as EDTA (ethylenediaminetetraacetic acid) and DTPA (diethylenetriaminepentaacetic acid) are employed to complex with insoluble metal ions like Fe^{3+} , Ca^{2+} , or Ba^{2+} , facilitating the dissolution of both acid-insoluble and passivated scale materials. These agents work under controlled pH conditions, minimizing corrosion risk while maintaining high metal-binding capacity. Simultaneously, anionic or nonionic surfactants are incorporated to disperse hydrocarbon-based residues and improve wettability, thereby enhancing the penetration and surface contact of the dissolving fluids with the scale matrix. These surfactants also help emulsify and solubilize the toluene-soluble organic matter, breaking down organic-mineral agglomerates and preventing redeposition.

Table 2 address solubility analysis revealed that the sample contains 0% water-soluble material, indicating complete insolubility in neutral aqueous conditions. The organic matter content was 7.27%, suggesting limited presence of hydrocarbons or degradable organics. The inorganic fraction was predominant, with 43.5% classified as acid-soluble material, likely consisting of carbonates or sulfates, and 49.23% as acid-insoluble, indicating a high concentration of chemically resistant minerals such as silica, barite, or refractory oxides. These results confirm that the sample is largely composed of inorganic compounds, with nearly half exhibiting strong resistance to acidic dissolution. This highlights the need for a tailored chemical approach, possibly involving chelating agents and mechanical assistance for effective remediation.

Table 2—summary of scale dissolution

Summary of Solubility Result	% weight
water soluble	0
Organic Matter	7.27
Inorganic matter as acid soluble material	43.5
Inorganic matter as acid insoluble material	49.23

The combined system is further optimized by adjusting fluid rheology, temperature compatibility, and contact time to ensure maximum dissolution efficiency while preserving the integrity of tubing materials. Pre-flush and post-flush stages using mutual solvents or water-based spacers are typically introduced to clean the tubing and remove any residual emulsions or metal-chelate complexes. The solubility test thus served as the foundation for tailoring this multi-component treatment strategy, ensuring full-spectrum coverage of the diverse scale constituents and enabling successful restoration of flow pathways in the production tubing. **Figure 3** shows the fresh scale analysis received from the field treated and analysed in order to evaluate its content and dissolve it.

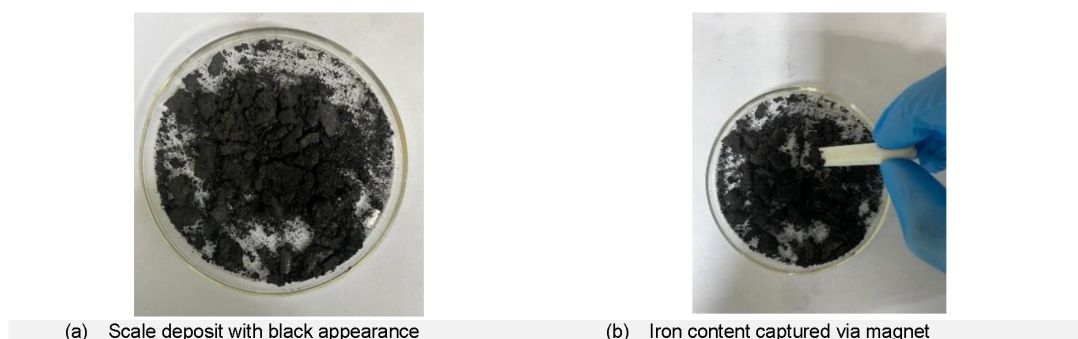


Figure 3—scale deposit received and treated in laboratory

A series of progressive solubility tests were conducted using hydrochloric acid (HCl) solutions of varying concentrations, ranging from 15% to 28%, to evaluate their effectiveness in dissolving the scale sample. Initial trials with 15% and 20% HCl showed limited reactivity, dissolving less than half of the deposit, indicating high acid resistance. However, when 28% HCl was applied under controlled conditions, it demonstrated significantly improved dissolution efficiency, achieving up to 41% weight reduction of the scale sample as shown in **figure 4**. The dissolution tests were visually documented and supported by experimental observations, including top, side, and bottom views of the reaction between the acid and scale within a transparent glass beaker, illustrating the extent of surface and structural breakdown. Following the soaking period, the residual solids were filtered and oven-dried at 105°C for 2 to 4 hours, ensuring complete removal of moisture and water content. This drying step was essential to accurately measure the remaining solid deposits and verify the true dissolution capacity of the acid. These results confirmed that only highly concentrated HCl solutions, such as 28%, are capable of effectively breaking down and solubilizing the scale material, emphasizing the need for aggressive acid systems in treatment formulations for such resistant deposits.

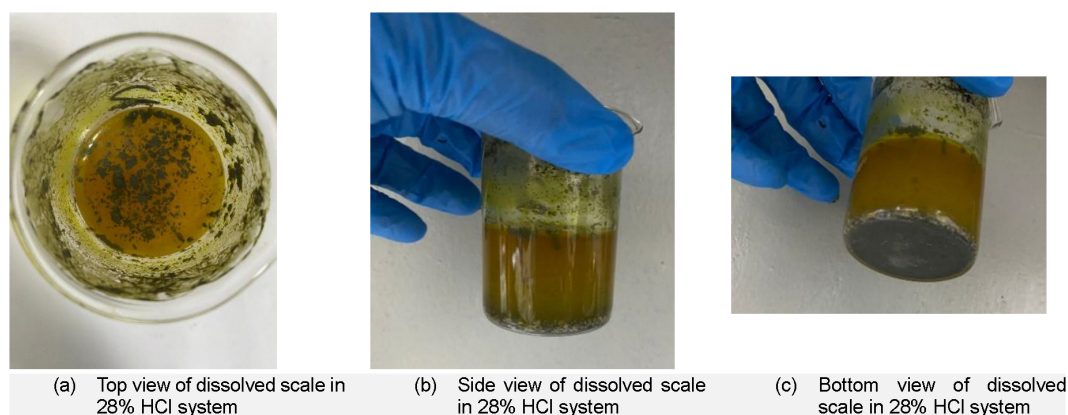


Figure 4—Scale dissolution via HCl system

The second phase involved systematic solubility and dissolution testing under simulated downhole conditions. Static soak tests were performed in jacketed glass batch reactors equipped with thermocouples and overhead stirrers to maintain constant agitation at temperatures up to 150°C. Additionally, oven machine were used for high-pressure testing and high temperature simulating reservoir-like confinement. A series of fluids were evaluated, including 15% hydrochloric acid (HCl), 28% hydrochloric acid 10% acetic acid, and various chelating agents notably EDTA (ethylenediaminetetraacetic acid) and DTPA (diethylenetriaminepentaacetic acid). Dissolution efficiency was quantified by gravimetric loss of scale samples. Tests revealed that HCl and acetic acid were ineffective against BaSO₄ and SrSO₄ due to their low solubility, and limited efficacy against iron-rich matrices.

Based on these findings, an EDTA-based chelating solution was formulated and further optimized through surfactant augmentation. A novel nano-surfactant system, composed of non-ionic and amphoteric molecules with particle sizes below 100 nm, was introduced to alter wettability from oil-wet to water-wet, reduce interfacial tension (IFT), and enhance penetration into hydrophobic, dense scale clusters. Thermal stability of the combined chelating/nano-surfactant fluid was validated in an aging cell oven under continuous thermal stress at 135–150°C for 48 hours.

The final treatment blend, composed of 39 wt% EDTA, 1 wt% nano-surfactant, and proprietary stabilizers, demonstrated superior dissolution efficiency: up to 65% mass reduction of BaSO₄ in 24 hours under HPHT conditions, and complete dissolution of iron oxide residues with no scale re-precipitation. The optimized formulation showed chemical resilience, flowability, and field handling compatibility, making it suitable for deployment via coiled tubing in high-temperature, high-pressure downhole environments.

A comprehensive series of solubility tests were performed using varying concentrations of chelating agents, specifically EDTA, to evaluate their effectiveness in dissolving acid-insoluble scale materials. Results demonstrated that increased saturation levels of the chelating agents significantly improved dissolution rates, confirming that the higher the chelation capacity, the greater the ability to bind and mobilize metal ions from the scale matrix. To further enhance the performance of the EDTA-based fluid system, particularly in terms of wettability and penetration into compacted scale structures, a nano-surfactant was introduced and uniformly blended with EDTA to form a stable, single-phase solution. This modified system improved the interaction between the fluid and scale surface, allowing for deeper infiltration and more efficient breakdown of deposits. As demonstrated in the supporting [figures 5](#), the scale deposits were visibly disintegrated, with spreading and progressive erosion clearly observed across treated surfaces. The combined action of EDTA and the nano-surfactant led to a 65% dissolution of the inorganic, acid-insoluble fraction, highlighting the formulation's enhanced capacity to target and remove resistant mineral scales under simulated downhole conditions.

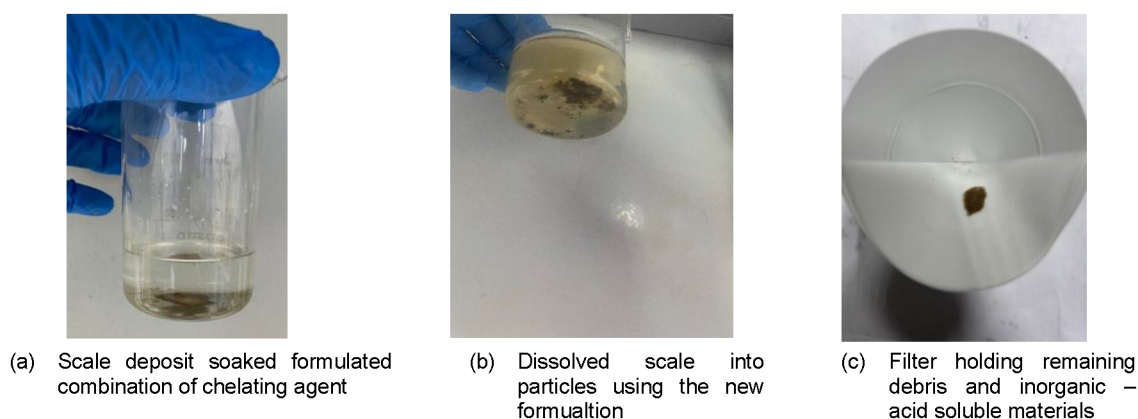


Figure 5—Scale dissolved in formulated chelating agent

Laboratory Result

In this study, a specially formulated and customized chelating agent system—comprising a synergistic blend of ethylenediaminetetraacetic acid (EDTA) and a nano-structured surfactant—was developed to overcome the persistent challenge of dissolving inorganic scale deposits under ambient, non-thermal conditions. The solution was designed to target complex multi-component scales typically encountered in oilfield environments, such as calcium carbonate (CaCO_3), barium sulfate (BaSO_4), iron oxides ($\text{Fe}_2\text{O}_3/\text{Fe}_3\text{O}_4$), and mixed silicates. A representative scale sample, sourced from a real field operation, was subjected to static soaking in the chelating solution at room temperature without any application of external heat. Over a 24-hour period, complete dissolution of the scale was achieved, as evidenced by the transition from a visibly heterogeneous solid-liquid interface to a fully transparent and homogeneous solution, with no visible particulates or sedimentation. This successful outcome is attributed to the high metal-ion complexation efficiency of EDTA, which disrupted the ionic lattice of the inorganic scale, coupled with the nano-surfactant's ability to enhance wettability, interfacial penetration, and molecular-level dispersion of the chelating agent across the solid surface.

The conclusion drawn from this work holds significant operational, technical, and economic implications. First, the dissolution of complex scale at ambient temperature proves that high-temperature assistance, which is often logistically challenging and cost-prohibitive. Second, the chelating system proved to be selective, efficient, and non-aggressive toward surrounding metallic infrastructure, preserving material integrity and minimizing corrosion risk, an essential advantage for applications in sensitive production and injection systems. Third, the complete dissolution within 24 hours under static conditions implies a high potential for real-time operational applications, such as batch treatments in wellbores, pipelines, separators, or surface tanks, without intensive mechanical intervention. Fourth, the chemical formulation uses readily available, environmentally compliant components, enhancing its applicability in regulatory-sensitive regions.

In summary, the tailored EDTA–nano-surfactant system offers a cost-effective, field-ready, and thermally independent solution for the remediation of inorganic scale in complex oilfield environments. Its demonstrated performance under mild conditions challenges conventional scale removal practices that rely heavily on harsh acids, high pressure, or elevated temperatures. This innovation opens the door for scalable deployment in both upstream and midstream sectors, while also serving as a model platform for future development of smart chelation-based remediation technologies.

Field application and implementation

The candidate well selected for the stimulation operation was a water injector well that had become completely clogged due to extensive scale deposition, resulting in zero injectivity prior to intervention. Previous well intervention attempts, including Wireline (W/L) diagnostics and acid stimulation, failed to establish communication or restore flow, indicating severe and compact blockage within the production string. The well was completed with a 3.5-inch production tubing, having an internal diameter (ID) of 2.99 inches, which significantly limited the space available for intervention tools and fluid placement. Diagnostic analysis and prior records confirmed that the tubing was obstructed from surface down to WLEG, presenting tubing length with continuous blockage. The severity and vertical extent of the obstruction posed significant operational challenges, as traditional mechanical and chemical methods were ineffective in penetrating or dissolving the buildup. This scenario necessitated the development and deployment of a high-performance fluid system, coupled with predictive modelling, to precisely target the scale and restore well injectivity.

Well description and details

The subject well is a vertical-to-Lateral water injection well, completed to a vertical depth of approximately 9,500 ft followed by an open-hole horizontal section. The completion consists of 3½" low carbon steel tubing (L-80 grade), with a minimum internal diameter of 2.69" at the no-go nipple. Over time, operational challenges emerged due to inorganic scale precipitation, primarily resulting from chemical incompatibility between injected water and formation fluids, compounded by pressure and temperature fluctuations. This led to progressive scale buildup across the tubing string, eventually causing complete blockage at the Surface-Controlled Subsurface Safety Valve (SCSSV).

The scale obstructed the safety valve internals, rendering it non-functional and blocking access for any downhole interventions via coiled tubing or wireline. With the restricted tubing ID and the obstruction location near the SCSSV, mechanical entry is no longer feasible. This situation not only impacts the well's injection efficiency but also violates safety and operational standards by compromising the SCSSV barrier.

Given the material sensitivity of L-80 tubing, which is susceptible to acid corrosion, a non-damaging chemical formulation was essential. Therefore, a specially formulated chelating solution based on EDTA (ethylenediaminetetraacetic acid) blended with a nano-surfactant-based additive was selected for treatment. This combination offers excellent scale-dissolving efficiency while ensuring minimal corrosivity toward low-carbon steel materials. The nano-surfactant enhances the wetting, penetration, and contact surface of the EDTA solution, allowing it to access micro-constricted areas within the scale matrix. This chemical system was selected specifically to restore flow paths through the SCSSV, dissolve the obstructive deposits, and preserve the metallurgical integrity of the tubing, eliminating the need for mechanical intervention while ensuring safe, controlled, and efficient scale removal in this constrained downhole environment.

Operational sequence implementation

The implementation of the scale removal operation was the culmination of extensive laboratory testing, strategic planning, and collaborative coordination between field personnel and engineering teams. The well had suffered from zero injectivity, and all conventional acid stimulation techniques had failed due to the extensive blockage caused by a combination of inorganic scales, predominantly calcium carbonate, iron oxides, barium sulfate, and other mineral deposits, severely accumulated along the tubing and near the perforation zone.

Initial discussions were held between the main operator's production and intervention engineers and the contractor's technical department, during which the need for a coiled tubing (CT) intervention was thoroughly assessed and approved. It was mutually agreed that the operation would be carried out in multiple phases, starting with chemical soaking using a customized chelating agent solution formulated based on laboratory results. This solution had been rigorously tested under controlled lab conditions, demonstrating its superior ability to dissolve mixed inorganic scales in less than 24 hours at ambient temperature without requiring thermal support, where the synergistic action of ethylenediaminetetraacetic acid (EDTA) as the primary complexing agent and a proprietary nano-surfactant designed to improve interfacial contact and permeability across the scale matrix.

The first day of field execution involved mobilizing the CT unit and preparing the wellsite. The intervention began by running the coiled tubing string into the well to reach top of the obstruction above SCSSSV. Upon tagging, the tubing was filled completely to surface with the pre-mixed chelating solution. Once the tubing capacity was entirely displaced and filled with the chelating solution, the string was pulled out of hole (POOH), and the well was left to soak under static conditions for a 24-hour period. This soaking period was chosen based on the laboratory experiments where complete dissolution of field-collected scale samples occurred within 24 hours without agitation or elevated temperature, confirming the capability of the formulation under field-representative conditions. During the soak period, intermittent surface monitoring and temperature logging confirmed that the solution remained chemically stable and well-retained.

On the next day, a hydro-mechanical intervention strategy was implemented, combining custom-formulated chemical soaking with high-pressure, rotational jetting using a precision-engineered self-rotating jetting and milling tool to address the severe scale blockage within the wellbore, particularly across the SCSSV and inner tubing walls. This tool, designed specifically for coiled tubing deployment, was customized to fit the restricted ID of the 3½" L-80 tubing, with a 2.125" OD body and progressively enlarged jetting heads reaching 2.5" bite size, enabling effective conveyance through constricted zones and direct engagement with scale deposits. The system operates via differential pressure, generating high RPM torque that powers four strategically placed high-pressure nozzles, two upward-facing and two downward-facing to create a balanced vortex flow pattern during pumping. This vortex facilitated the dislodgement of loosened scale particles and their upward transport within the annular flow, effectively preventing re-deposition along the wellbore, and simultaneously attack scale bridges from both directions, optimizing flow contact and mechanical erosion. The conical carbide-dressed head at the bottom provides an abrasive milling surface designed not only to break hard, mineralized obstructions but also to prevent cuttings or loosened scale from accumulating on the tool face, thereby mitigating the risk of stuck pipe or tool immobilization as demonstrated in figure 6.

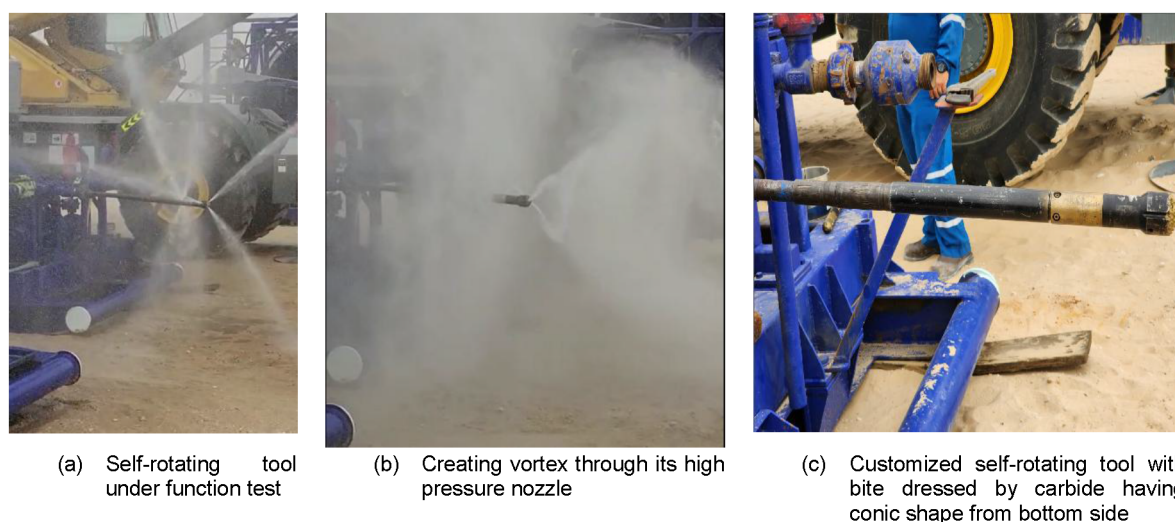


Figure 6—Self-rotating jetting tool creating vortex

Post SCSSS-V retrieval by Wireline, same procedure was repeated, the CT string was run to a deeper depth, which corresponded to the lower bound of the suspected scale plug. Once at depth, a new batch of the formulated chelating solution, identical in formulation to the one previously used, was pumped from bottom to surface, ensuring a full tubing displacement with fresh reactive fluid. Again, the string was pulled out of hole and the well was left to soak for an additional 24-hour period. The rationale behind this dual-stage soaking process was to maximize chemical contact time over different zones of the tubing and to ensure that the chelating solution had sufficient opportunity to penetrate microchannels, dissolve internal and surface-bound deposits, and reduce scale thickness prior to any mechanical intervention. The soaking also allowed for significant weakening and chemical disintegration of the scale structure, which was expected to enhance the performance of mechanical removal tools in the subsequent phase [7].

By the third day, the operation transitioned into the mechanical descaling phase. The self-rotating, pressure-activated jetting tool was rigged up and run into the well on coiled tubing down to Wireline Entry Guide (WLEG). The jetting effect allowing the tool to target and break apart any residual scale adhering to the tubing inner diameter or lodged in tight spots. The selection of this tool was based on its compatibility with the CT system, its operational reliability under field conditions, and its proven performance in past interventions involving difficult scale profiles.

High-pressure acid jetting was initiated once the rotating tool reached target depth, and multiple passes were made while gradually pulling the tool out of hole to target residual iron-based scale deposits that were resistant to the initial chelating treatment. This acid wash employed an operator-approved, corrosion-inhibited acid formulation specifically designed to dissolve iron oxides and sulfides without compromising the integrity of the tubing or downhole equipment. After 4-Hours soaking time, the inhibited acid was circulated under the same operational conditions, utilizing the jetting tool's multi-directional jets and rotating action to maximize acid contact with the scale surface and enhance dissolution kinetics. The dynamic fluid action, combined with the pre-softened state of the scale due to chelating soak, enabled efficient removal of remaining deposits. This targeted acid treatment successfully dissolved stubborn iron-based scale deposits. Visual and pressure data confirmed improved fluid circulation and a gradual return of injectivity, in addition, Wireline run confirmed full-bore tubing clearance.

The mechanical intervention successfully cleared all obstructed tubing sections and ensured clean flow paths for subsequent well operations. No fallback material was observed during post-job wellhead sampling, indicating complete removal of the targeted scale. The combined chemical-mechanical approach proved highly effective, validating laboratory predictions and demonstrating the operational feasibility of using a low-temperature chelating soak followed by precision mechanical Acid-jetting to restore well integrity.

The sequential combination of chemical slugs, starting with the hydrochemical chelating soak, followed by mechanical jetting, and finalized by the inhibited acid wash, proved sufficient to fully overcome the diverse scale deposition that had rendered the well completely blocked and inactive for an extended period. The integrated chemical-mechanical-chemical sequence restored well injectivity and flow capacity allowing the full restoration of the well's mechanical integrity, safety systems, transforming a non-operational asset back into a productive state.

The success of this multi-day operation provides a replicable blueprint for tackling similar challenges across other wells with complex scale deposition under non-reactive or low-injectivity conditions.

Conclusion

The intervention successfully addressed a complex and high-impact integrity challenge in a critical water injection well where severe scale deposition had completely obstructed the completion string, including the Surface-Controlled Subsurface Safety Valve (SCSSV), and extended down to the Wireline Entry Guide (WLEG). This blockage not only compromised the well's injectivity and flow assurance but also rendered the SCSSV inoperative, eliminating the positive safety barrier, and posed significant risks to well control and intervention capabilities. A thorough diagnostic phase was conducted, including in-depth scale characterization, which revealed a composition resistant to conventional solvents. Based on these findings, a specialized chelating treatment was formulated, combining EDTA-based chemistry with a tailored nanosurfactant package, ensuring optimal dissolution without compromising the integrity of the L-80 low-carbon steel tubing.

To deliver the chemical effectively and mechanically remove hardened scale, a comprehensive strategy was implemented using a custom-designed, self-rotating jetting and milling tool. Key highlights of the operation include:

- **Customized Tool Design:**
 - *The tool featured a compact outer diameter (OD) with interchangeable jetting heads.*
 - *Tailored to fit the minimum internal diameter of the 3 ½" tubing (2.69"), enabling safe navigation through tight completion profiles.*
- **Dual-Action Cleaning Mechanism:**

- *High-pressure radial jetting through four strategically placed nozzles (two upward, two downward).*
- *Simultaneous torque generation and mechanical abrasion to erode, loosen, and dislodge scale bridges and deposits.*
- **Multi-Stage Intervention Strategy:**
 - *Deepen tool penetration into the scale-filled tubing.*
 - *Increase torque and fluid energy at each stage.*
 - *Improve chemical treatment contact time and coverage.*
- **Optimized Chemical Delivery:**
 - *Strategic pumping and soaking sequences allowed the formulated chelating agent to act efficiently across the entire obstructed section.*
 - *Tool's design prevented scale accumulation on the bit, reducing stuck-pipe risk.*
- **Restoration of Well Integrity and Functionality:**
 - *Complete restoration of tubing access down to the Wireline Entry Guide (WLEG).*
 - *SCSSV fully cleaned, revalidated, and functionally restored, re-establishing the safety barrier.*
 - *Well injectivity successfully regained, enabling full operational recovery.*

This integrated hydro-mechanical and chemical approach proved critical in overcoming severe wellbore obstruction and restoring injection capacity without invasive interventions.

This success was the result of a collaborative effort between the operator's senior engineering team and the contractor's technical department, with a shared objective of restoring well functionality, securing mechanical integrity, and improving return on investment by avoiding costly recompletions or sidetracks. The intervention sets a new benchmark for tackling severe scale-related failures in constrained geometries using intelligent chemical and mechanical integration.

Reference

1. Yousuf M. Al Rawahi, Feroz Shaik. Studies on Scale Deposition in Oil Industries & Their Control, *International Journal for Innovative Research in Science & Technology*, Volume 3, Issue 12, May 2017.
2. Asif Ali, Ning Zhang, Rafael Mattos Dos Santos. Mineral Characterization Using Scanning Electron Microscopy (SEM): A Review of the Fundamentals, Advancements, and Research Directions, *Applied Sciences*, November 2023.
3. Tariq Almubarak ORCID logo, Jun Hong Ng, Raja Ramanathan and Hisham A. Nasr-El-Din. From initial treatment design to final disposal of chelating agents: a review of corrosion and degradation mechanisms, 12th January 2022.
4. Timur Ildarovich Yunusov, Lyutsia Faritovna Davletshina, Dmitriy Nikolaevich Klimov, Lyubov Abdulaevna Magadova and Mikhail Alexandrovich Silin. Study of Wettability Alteration of Hydrophobic Carbonate Rock by Surfactant-Containing Chelating Agent Solutions, MDPI, 26 August 2023.
5. Abass A. Olajire. A review of oilfield scale management technology for oil and gas production, *Journal of Petroleum Science and Engineering*, Volume 135, November 2015, Pages 723–737

6. Mir Salar Hamzeh, Mahsa Parhizgar Keradeh, Seyyed Alireza Tabatabaei-Nezhad. Exploring the Impact of EDTA and DTPA Chelating Agents on Rock-Fluid Interactions, *Fluid-Fluid Interactions and Fluid Flow in Carbonate Cores Under HP-HT Reservoir Conditions*, SSRN, 18 Aug 2024.
7. Amjed Hassan, Mohamed Mahmoud, Badr S. Bageri, Murtada Saleh Aljawad, Muhammad Shahzad Kamal, Assad A. Barri, Ibelwaleed A. Hussein. Applications of Chelating Agents in the Upstream Oil and Gas Industry: A Review, ACS publications, November 23, 2000.